data through neighbor-to-neighbor pathways to preserve command-and-control capability. This enhances resilience by preventing the loss of a single communication relay from compromising the entire system.

Operational testing has demonstrated that quantum-resilient communication enables drones to operate reliably in GPS-denied, signal-jammed, or cyber-contested airspace, which is increasingly common in modern military theatres. Such architecture is fundamental for loyal wingman UAVs, swarm drones, reconnaissance assets, and long-range autonomous strike platforms that must maintain mission integrity even under electronic warfare pressure.

While the technology is promising, challenges remain, including the miniaturization of QKD hardware, ensuring energy efficiency for onboard processors, and establishing interoperability standards across allied defense systems. Nonetheless, the US quantum-resilient UAV communication initiative marks a critical step toward securing next-generation autonomous air operations, positioning the architecture as a foundational layer for future air combat networks, unmanned traffic management, and global drone mobility systems.

9.8.11 European Union: Quantum key distribution for drone swarm networks

Within the European Union, research into drone swarm communication security has increasingly focused on the deployment of quantum key distribution (QKD) to enable secure, tamper-evident communication among autonomous UAV networks. Supported by initiatives such as the EU Quantum Flagship Program, defense research agencies, and national laboratories in France, Germany, and the Netherlands, these efforts aim to protect swarm coordination links from interception or signal spoofing – particularly in future operational environments where adversaries may